

Grade 1



**LUNAR
ASSESSMENTS**

12 STEPS TO PROGRESSIVE LEARNING

STUDY PACK - 7

mathlab
INSTITUTE

MaRRS
REDISCOVER
CHALLENGE | DISCOVER | LEARN

PREFACE

Mathematics is not just about numbers and formulas—it is a way of thinking, reasoning, and solving problems. The Lunar Skill Assessment is designed with this vision in mind, to help students build a strong and lasting foundation in mathematics through continuous and structured practice.

Each month, students receive a study pack that includes guided lessons, practice problems, mock tests, and actual assessments for four weeks. These sessions are aligned with a well-planned syllabus, ensuring that learning happens in a systematic and consistent manner. By engaging with this monthly content week after week, students not only master mathematical concepts but also develop confidence and discipline in their studies.

We believe that mathematics is best learned as a journey, where each step builds upon the previous one. With dedication and consistency, this program will empower students to enjoy the process of learning, sharpen their skills, and excel in mathematics with confidence.

RECAP: Study Pack 6

(Addition within 20)

1. What is the sum of 12 and 3?
(A) 15
(B) 14
(C) 16
(D) 13
2. If you have 10 chocolates and your friend gives you 7 more, how many do you have now?
(A) 10
(B) 17
(C) 18
(D) 20
3. Which of these equals 20?
(A) $10 + 5$
(B) $15 + 4$
(C) $10 + 10$
(D) $12 + 7$
4. $14 + 5 = \text{-----}$
(A) 19
(B) 18
(C) 17
(D) 20
5. What is 11 plus 2?
(A) 12
(B) 14
(C) 15
(D) 13
6. Find the missing number: $8 + \text{----} = 15$.
(A) 5
(B) 7

(C) 8

(D) 6

7. Look at the image below. How many stars are there in total?



(A) 12

(B) 13

(C) 14

(D) 15

8. $13 + 4$ is the same as:

(A) 17

(B) 16

(C) 15

(D) 18

9. If there are 9 red balls and 9 blue balls, what is the total?

(A) 17

(B) 18

(C) 19

(D) 20

10. Which sum is greater than 15?

(A) $10 + 2$

(B) $11 + 3$

(C) $12 + 1$

(D) $9 + 8$

RECAP: Study Pack 6

Solution Manual

1. (A) 15
Start at 12 and count forward 3 steps: 13, 14, 15.
2. (B) 17
10 and 7 more makes 17.
3. (C) $10 + 10$
Adding two tens makes twenty ($10 + 10 = 20$).
4. (A) 19
 $14 + 5 = 19$. You can use your fingers to count 5 more from 14.
5. (D) 13
 $11 + 2 = 13$.
6. (B) 7
Count on from 8 to reach 15: 9, 10, 11, 12, 13, 14, 15. That is 7 steps.
7. (C) 14
Counting the two groups of stars in the image gives a total of 14.
8. (A) 17
 $13 + 4 = 17$.
9. (B) 18
Double 9 is 18 ($9 + 9 = 18$).
10. (D) $9 + 8$
 $9 + 8 = 17$, which is greater than 15.

Study Pack 7: Mental Addition

PREPARATORY MATERIAL

Learning Outcomes

By the end of this lesson, learners will be able to:

- Add numbers up to 10 mentally without using fingers or paper.
- Use the "counting on" strategy to solve addition problems.
- Quickly add 0, 1, and 2 to any number.
- Identify doubles (like $2+2$, $3+3$) to speed up calculations.
- Solve simple word problems using mental math.
- Understand number bonds that make 10.

Key Facts to Remember

- Mental math means doing math in your **head**.
- When we add **0** to a number, the number stays the same!
- Adding **1** is just finding the "next number."
- To count on, start with the **bigger** number and count the smaller number forward.
- Number bonds are pairs of numbers that fit together like a puzzle to make a bigger number.

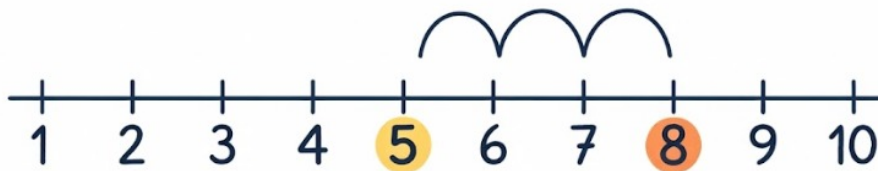
Core Concepts

1. What is Mental Addition?

Mental addition is the ability to calculate the sum of numbers in your mind. Instead of writing it down or using your fingers, you use "tricks" or strategies to find the answer quickly.

The Counting On Strategy

This is the most helpful trick for beginners. *Example:* To find $5 + 3$: 1. Start with the bigger number: **5**. 2. Count forward 3 steps in your head: **6, 7, 8**. 3. The answer is **8**.



Think of a number line in your head and "jump" forward!

2. Fast Addition with 0, 1, and 2

Some numbers are so easy to add that we can do it in a blink!

Adding Zero (The Hero)

Adding zero doesn't change anything. $7 + 0 = 7$ $10 + 0 = 10$

Adding One (The Next Neighbor)

Adding one is simply the next number in the counting sequence. $4 + 1 = 5$ $9 + 1 = 10$

Adding Two (The Skipper)

Adding two is like skipping one number. $3 + 2 = 5$ (Think: 3... skip 4... 5!)

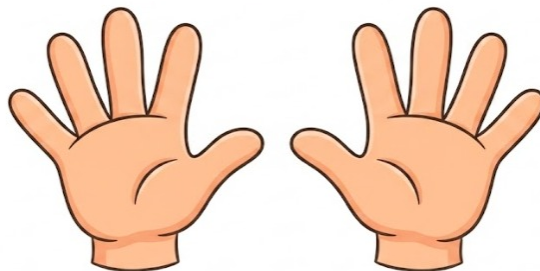


Adding one gives the next neighbour.

3. Learning Your Doubles

Doubles are when you add the same number to itself. Learning these by heart makes you a math wizard!

- $1 + 1 = 2$
- $2 + 2 = 4$
- $3 + 3 = 6$
- $4 + 4 = 8$
- $5 + 5 = 10$



Using your two hands to show $5 + 5$ is a great way to remember doubles!

4. Number Bonds to 10

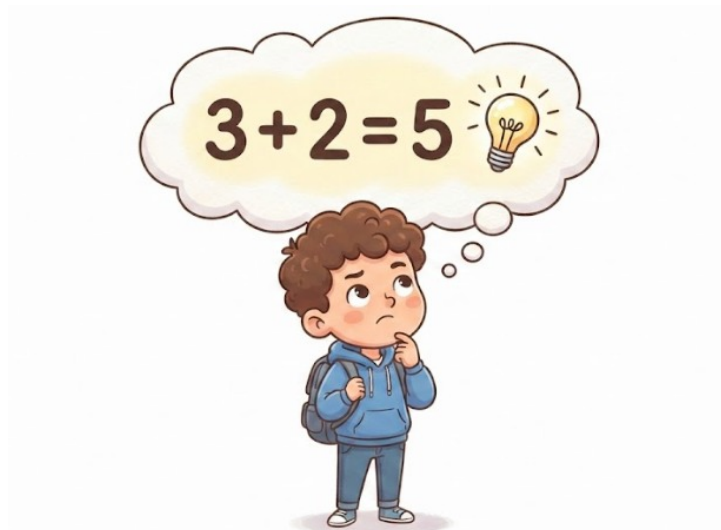
Knowing which two numbers make 10 is a very important mental math skill. These pairs are "best friends":

- 1 and 9
- 2 and 8

- 3 and 7
- 4 and 6
- 5 and 5

5. Tips for Better Mental Math

1. Visualize: Close your eyes and see the numbers in your head.
2. Practice Daily: The more you do it, the faster you get!
3. Use Objects: Start by counting real objects like apples or toy cars, then try to imagine them.



Always keep a "math bubble" in your head!

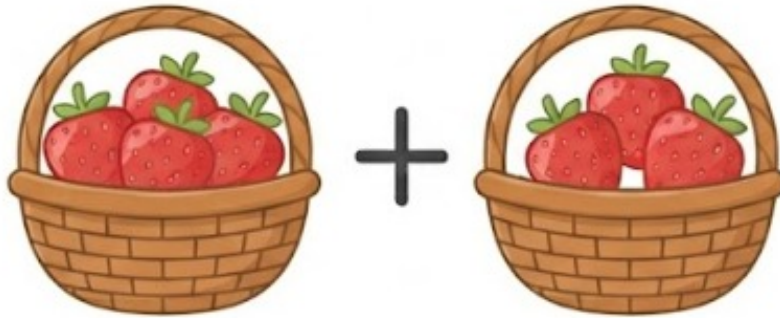
Practice Questions

DAY 01

1. What is $4 + 2$ calculated mentally?

- (A) 6
- (B) 5
- (C) 7
- (D) 4

2. Look at the picture. How many Strawberries are there in total?



- (A) 5
- (B) 7
- (C) 6
- (D) 8

3. Count on from 7: what is $7 + 3$?

- (A) 9
- (B) 8
- (C) 10
- (D) 11

4. What is $8 + 0$?

- (A) 8
- (B) 0
- (C) 9
- (D) 80

5. Double of 3 is:

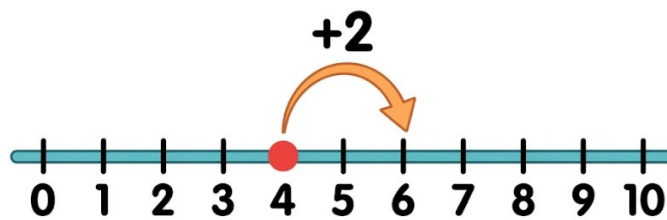
- (A) 5
- (B) 3
- (C) 9
- (D) 6

DAY 02

6. What is the next number after 9? ($9 + 1$)

- (A) 8
- (B) 10
- (C) 11
- (D) 12

7. Which addition sentence matches the jumps on this number line?



- (A) $2 + 2 = 4$
- (B) $3 + 1 = 4$
- (C) $4 + 2 = 6$
- (D) $5 + 1 = 6$

8. $5 + 2 = \text{-----}$

- (A) 7
- (B) 6
- (C) 8
- (D) 5

9. If you have 6 marbles and find 2 more, how many do you have?

- (A) 7
- (B) 8
- (C) 9
- (D) 6

10. Which number makes 10 when added to 5?

- (A) 4
- (B) 6
- (C) 3
- (D) 5

DAY 03

11. What is $3 + 4$ mentally?

- (A) 6
- (B) 8
- (C) 7
- (D) 5

12. Start with 6 and count on 1. What is the sum?

- (A) 7
- (B) 6
- (C) 8
- (D) 5

13. $2 + 2 + 2 = \text{-----}$

- (A) 4
- (B) 6
- (C) 8
- (D) 5

14. Look at the dice. What is the total number of dots?



- (A) 5
- (B) 6
- (C) 7
- (D) 8

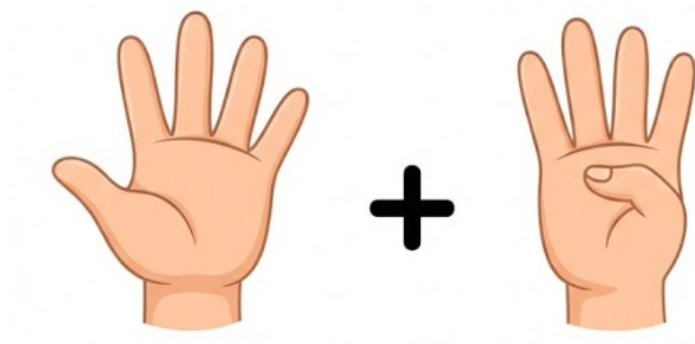
15. What is 0 added to 10?

- (A) 0
- (B) 1
- (C) 11
- (D) 10

DAY 04

16. Adding 1 to any number gives the:
- (A) Next number
 - (B) Previous number
 - (C) Same number
 - (D) Zero
17. $4 + 4 = \text{-----}$
- (A) 7
 - (B) 8
 - (C) 9
 - (D) 10
18. Which two numbers make 5?
- (A) 1 and 3
 - (B) 2 and 2
 - (C) 2 and 3
 - (D) 4 and 2
19. If you have 1 cat and get 1 more, you have:
- (A) 2 cats
 - (B) 1 cat
 - (C) 3 cats
 - (D) 0 cats

20. How many fingers are being held up in total?



- (A) 6
- (B) 8
- (C) 7
- (D) 9

DAY 05

21. Mentally solve: $5 + 1 + 1$.

- (A) 6
- (B) 7
- (C) 8
- (D) 5

22. Which is the same as $3 + 2$?

- (A) 4
- (B) 6
- (C) 5
- (D) 3

23. How many balloons are there in this bunch?



- (A) 6
- (B) 7
- (C) 8
- (D) 9

24. $1 + 9 = \text{-----}$

- (A) 10
- (B) 9
- (C) 11
- (D) 8

25. If I have 7 and I add 2, I get:

- (A) 8
- (B) 7
- (C) 10
- (D) 9

Student Evaluation Sheet

Study Pack 7 : Mental Addition

Q	Options Layout			
1.	☐ A	☐ B	☐ C	☐ D
2.	☐ A	☐ B	☐ C	☐ D
3.	☐ A	☐ B	☐ C	☐ D
4.	☐ A	☐ B	☐ C	☐ D
5.	☐ A	☐ B	☐ C	☐ D
6.	☐ A	☐ B	☐ C	☐ D
7.	☐ A	☐ B	☐ C	☐ D
8.	☐ A	☐ B	☐ C	☐ D
9.	☐ A	☐ B	☐ C	☐ D
10.	☐ A	☐ B	☐ C	☐ D
11.	☐ A	☐ B	☐ C	☐ D
12.	☐ A	☐ B	☐ C	☐ D
13.	☐ A	☐ B	☐ C	☐ D
14.	☐ A	☐ B	☐ C	☐ D
15.	☐ A	☐ B	☐ C	☐ D
16.	☐ A	☐ B	☐ C	☐ D
17.	☐ A	☐ B	☐ C	☐ D
18.	☐ A	☐ B	☐ C	☐ D
19.	☐ A	☐ B	☐ C	☐ D
20.	☐ A	☐ B	☐ C	☐ D
21.	☐ A	☐ B	☐ C	☐ D
22.	☐ A	☐ B	☐ C	☐ D
23.	☐ A	☐ B	☐ C	☐ D
24.	☐ A	☐ B	☐ C	☐ D
25.	☐ A	☐ B	☐ C	☐ D

Solution Manual

1. (A) 6

Start with 4 and count forward 2 steps: 5, 6.

2. (B) 7

The image shows a group of 4 strawberries and another group of 3 strawberries.

$$4 + 3 = 7.$$

3. (C) 10

Start with 7 and count 3 more: 8, 9, 10.

4. (A) 8

Adding 0 to any number leaves the number unchanged.

5. (D) 6

Double 3 means $3 + 3 = 6$.

6. (B) 10

Adding 1 is just the next number in sequence. 10 follows 9.

7. (C) $4 + 2 = 6$

The number line starts at 4 and makes a jump of 2 to land on 6.

8. (A) 7

$$5 + 2 = 7.$$

9. (B) 8

6 and 2 more makes 8.

10. (D) 5

The number bond for 10 with 5 is 5 ($5 + 5 = 10$).

11. (C) 7

$$3 + 4 = 7.$$

12. (A) 7

$$6 + 1 = 7.$$

13. (B) 6

Adding 2 three times: $2 + 2 = 4$, and $4 + 2 = 6$.

14. (C) 7

One die shows 4 dots and the other shows 3 dots. $4 + 3 = 7$.

15. (D) 10

$$10 + 0 = 10.$$

16. (A) Next number

This is the rule of adding 1.

17. (B) 8

$$4 + 4 = 8 \text{ (Double 4).}$$

18. (C) 2 and 3

$2 + 3 = 5$. The other options do not equal 5.

19. (A) 2 cats

$$1 + 1 = 2.$$

20. (D) 9

One hand shows 5 fingers and the other shows 4. $5 + 4 = 9$.

21. (B) 7

$$5 + 1 = 6, \text{ and } 6 + 1 = 7.$$

22. (C) 5

$$3 + 2 = 5.$$

23. (C) 8

Counting the balloons in the bunch gives a total of 8.

24. (A) 10

$$1 + 9 = 10 \text{ (Number bond to 10).}$$

25. (D) 9

$$7 + 2 = 9.$$